scPrediXcan integrates deep learning methods and single-cell data into a cell-type-specific transcriptome-wide association study framework

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Background

Transcriptome-wide association study(TWAS)

DNA $\rightarrow$ RNA(expression) $\rightarrow$ Trait(disease)

- **Definition**: TWAS is a method that tests the association between Gene Expression and a Trait (Disease).
- **The Goal**: To prioritize "Causal Genes." It tells us which gene is driving the disease and in what direction (e.g., "High expression of Gene A causes Disease B").
- **Dataset**: tissue-level gene expression data from a reference panel of individuals.

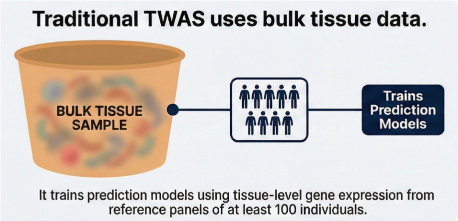


Figure: Traditional TWAS

The challenge of TWAS

- **Mismatch** between expression and disease-relevant cell types or states → single-cell data is needed
- **Unavailable** on the scale

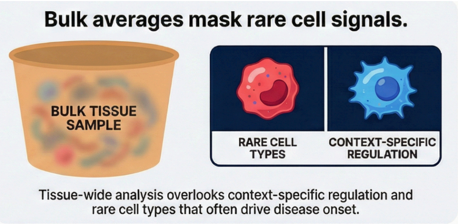


Figure: Challenge 1

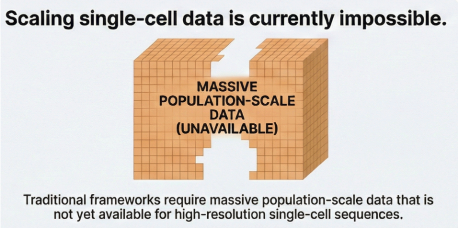


Figure: Challenge 2

→Which motivates a cell-type-specific gene expression prediction model that leverages pseudo-bulk scRNA-seq data

Method

scPrediXcan: First-step(ctPred)

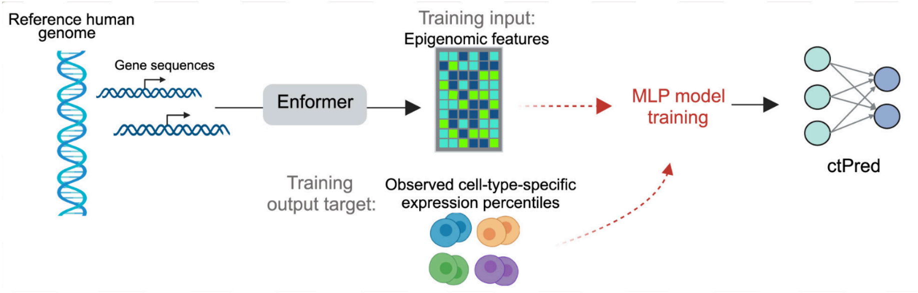


Figure: scPrediXcan: First-step

1. Input DNA sequence
2. Feature-extract using **Enformer**(pre-trained sequence-to-epigenomics model)
3. trained on single-cell RNA-seq data using four-layer **multi-layer perceptron (MLP)**
4. It learns to map those epigenetic features to the actual gene expression level for a specific cell type.

Enformer

- **Goal**: Predict gene expression and chromatin states from DNA sequences
- Standard models (CNNs like DeepSEA or Basenji) had limited "vision." They couldn't "see" far enough to catch these distant connections.

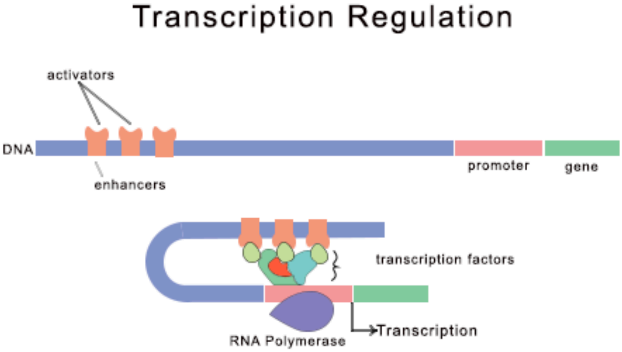
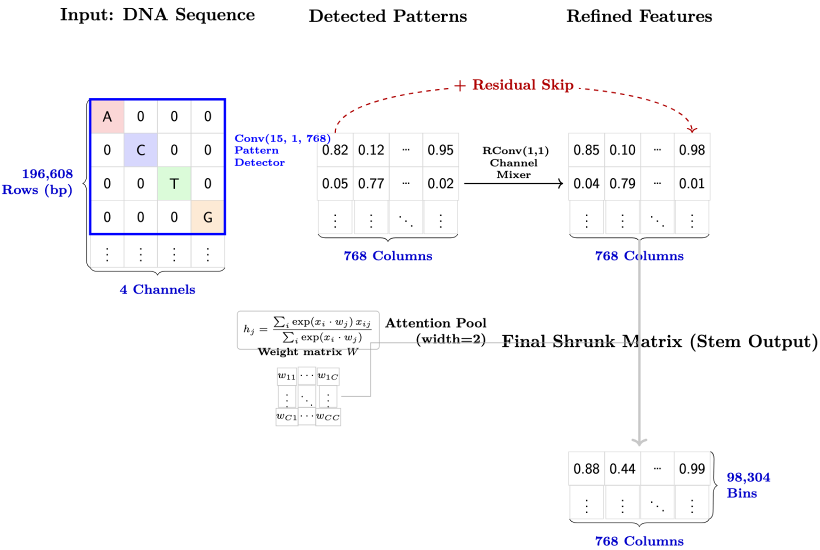
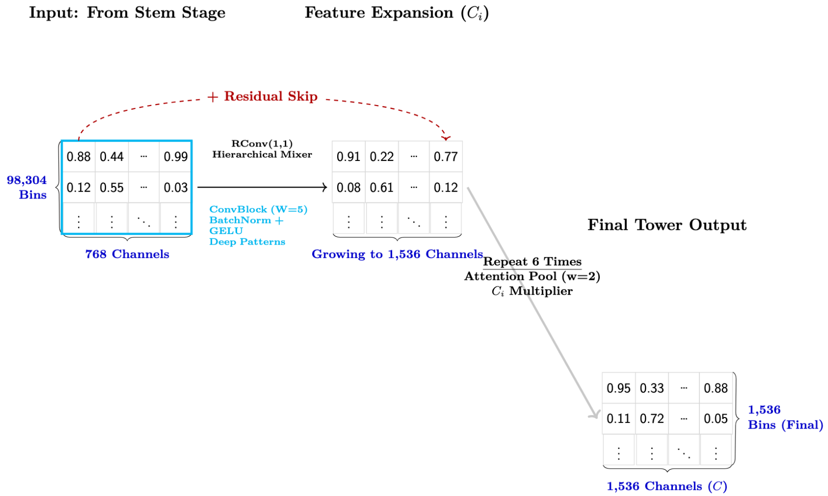


Figure: Transcription Regulation

Enformer's Architecture: Stem



Enformer's Architecture: Conv Tower

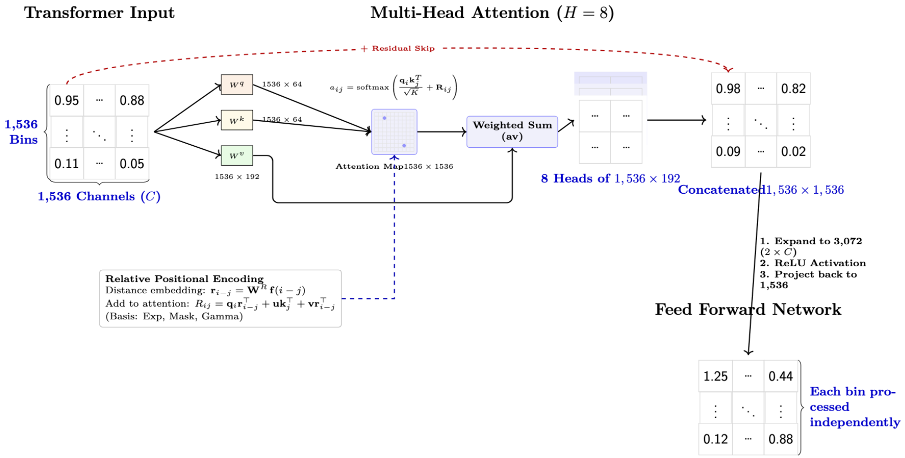


Conv Tower Logic:
1. Scan slightly wider ($W = 5$)
2. Expand features ($768 \to 1536$)
3. Shrink space ($98k \to 1.5k$)

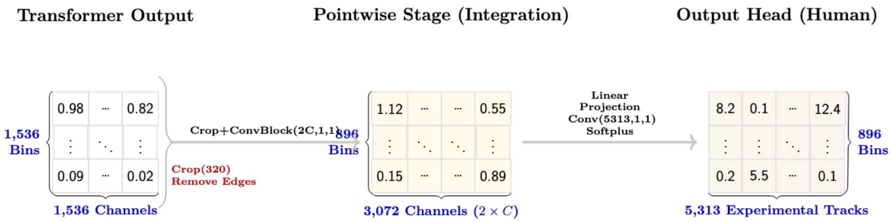
Enformer's Attention Mechanism

- **Model**: Transformer-based model which consists attention layers. For each position in the sequence, the model will calculate a weighted sum across the representations of all other positions in the sequence.
- **The "Attention" Mechanism**: Just like ChatGPT knows that a word at the start of a sentence connects to a word at the end, Enformer uses "Self-Attention" to understand how distant parts of DNA talk to each other.
- Enformer use a multi-head attention with $H = 8$ heads.

Enformer's Architecture: Multi-Head Attention



Enformer's Architecture: Pointwise + Output Head



Biological Results:
Positive values only (counts/peaks).
Every row = 128bp location.
Every column = A specific tissue/assay.

Enformer

- **Input**: 196,608-bp sequence of DNA as input
- **Output**: 5,313 × 896 output matrix(5,313 epigenomic features:For the human genome, each example contains 2,131 transcription factor (TF) chromatin immunoprecipitation and sequencing (ChIP—seq), 1,860 histone modification ChIP—seq, 684 DNase-seq or ATAC-seq, and 638 CAGE tracks)
- **Result**: The model has not only learned about the role of tissue-specific enhancers and promoters, but also about insulator elements and their role in inhibiting information flow between genomic compartments.

Train the ctPred

1. Three scRNA-seq datasets:OneK1K dataset, a T2D islet dataset, the Tabula Sapiens dataset
2. Evaluated the models'performance by calculating Pearson correlations between predicted and observed pseudobulk gene expression percentiles across the test sets

ctPred's Performance

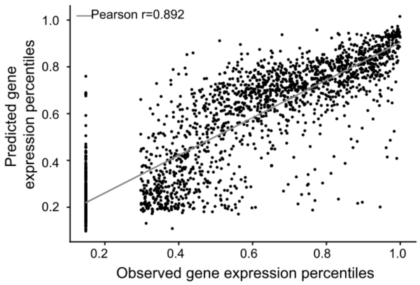


Figure: predicted gene expression percentiles and observed gene expression percentiles for each gene in a T-cell

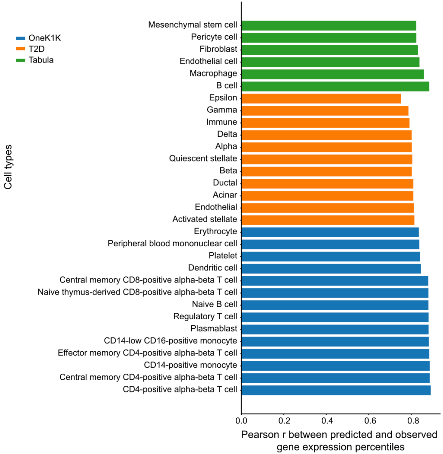


Figure: Pearson correlations between predicted and observed gene expression percentiles

Impact of epigenomic features on ctPred's model predictions

Shapley value analysis on T-cell:

1. Among the top 10 most impactful features, seven were immune related
2. T cell-specific features exhibited higher Shapley additive explanation compared to non-T cell features.

→ ctPred prioritizes cell-type-specific epigenomic features to enhance the accuracy of cell-type-specific gene expression predictions.

Drawback of the first step

1. The **computational expense** of such predictions for TWAS association tests across large GWAS cohorts remains high(predicting gene expression for 500 individuals across 20,000 genes would require ∼2,700 graphics processing unit (GPU)-hours.)
2. Access to **individual-level** data is limited.

→ Transforming the deep learning model into a linear form

scPrediXcan: Second-step(ℓ-ctPred)

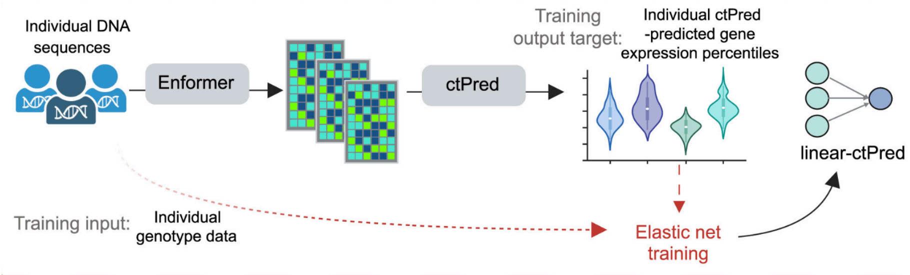


Figure: scPrediXcan: Second-step

1. Input the individual's DNA sequence, processes it through Enformer + ctPred
2. Train the the AI-Predicted Expression on individual genotype data to get set of weights for SNPs

They do not use the real measured expression here. They use the expression predicted by the AI. This effectively teaches the next model "how the AI thinks."

ℓ-ctPred approximate ctPred

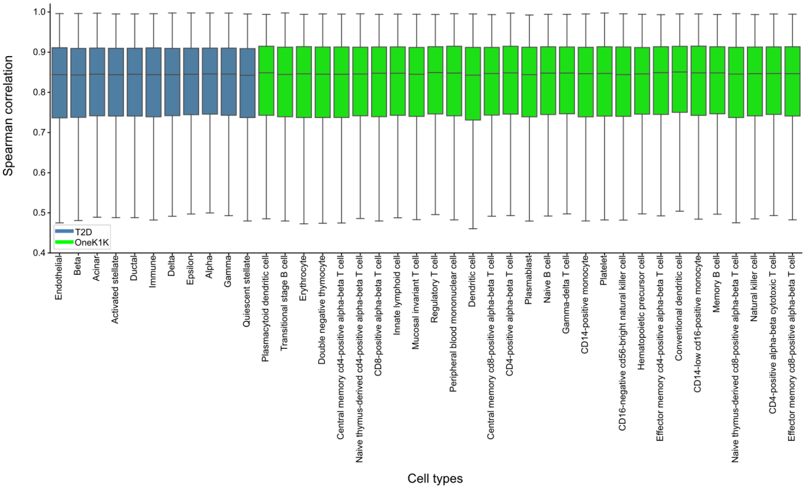
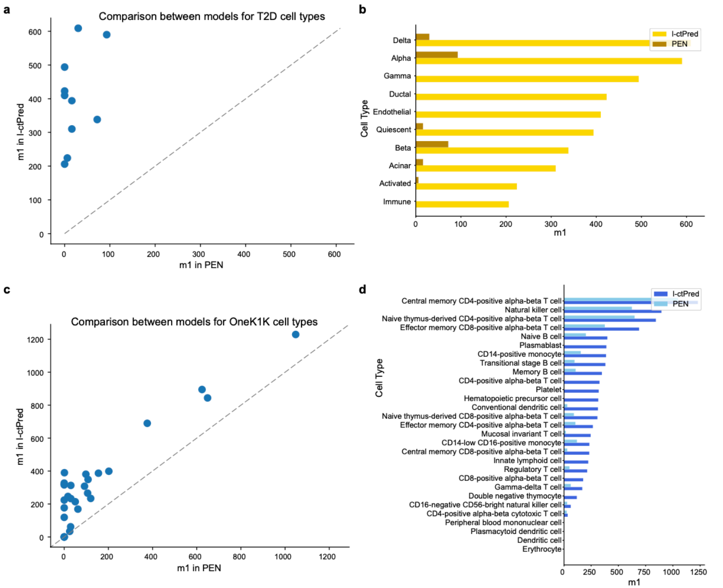


Figure: ℓ-ctPred approximate ctPred

ℓ-ctPred's Performance



scPrediXcan: Third-step(scPrediXcan)

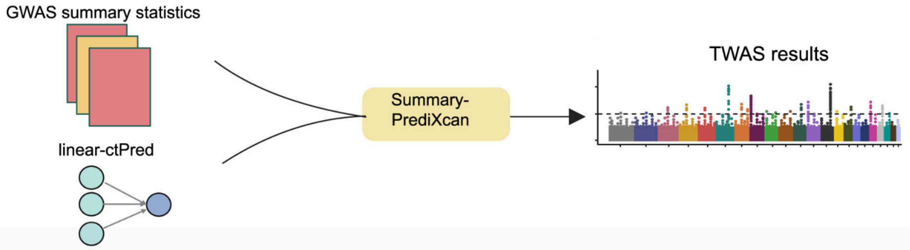


Figure: scPrediXcan: Third-step

1. Input the SNP weights from ℓ-ctPred and the GWAS summary statistics
2. The Z score: $Z_g = \sum_{l \in \text{scPred}} \omega_{g,l} \frac{\hat{\sigma}_l}{\hat{\sigma}_g} \frac{\hat{\beta}_l}{se(\hat{\beta}_l)}$
3. A two-tailed p-value can be calculated from the Z score using a normal distribution.

scPrediXcan's Performance

The comparisions:

1. a cell-type-specific canonical TWAS method (TWAS-pseudobulk) that uses the same scRNA-seq pseudobulk data as scPrediXcan (i.e., PEN predictors)
2. a tissue-level canonical TWAS method (TWAS-bulk) that relies on bulk RNA- seq data
3. other gene prioritization methods for T2D (STAR Methods)

scPrediXcan's Performance

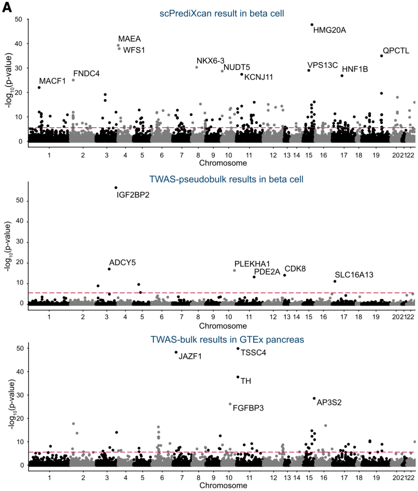


Figure: scPrediXcan identify more causal genes than other methods

scPrediXcan's Performance

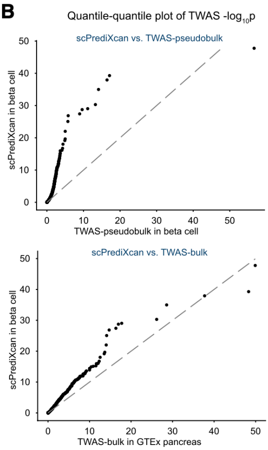


Figure: QQ plot of the statistical significance

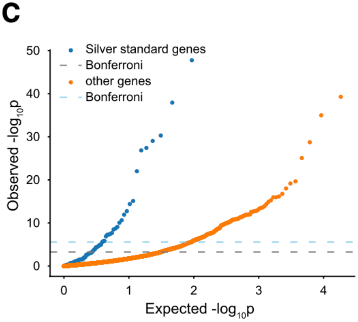


Figure: silver-standard genes has lower p-value

scPrediXcan's Performance

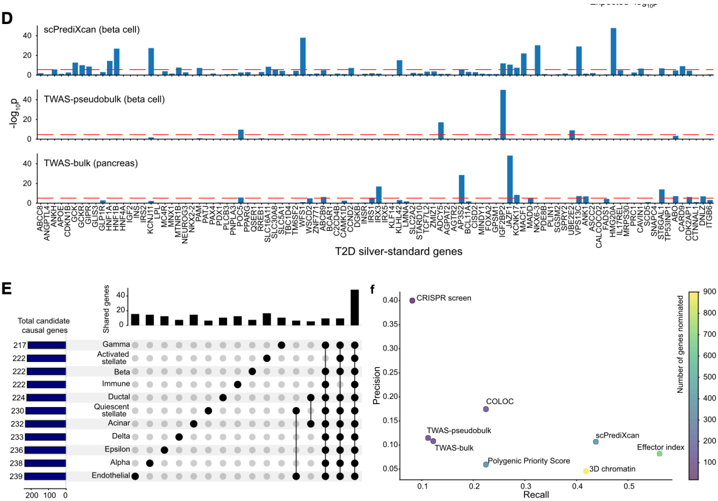


Figure: scPrediXcan vs other methods

scPrediXcan's Performance

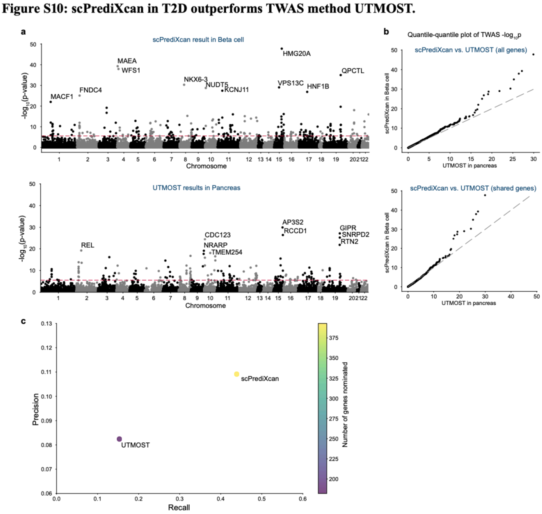


Figure: scPrediXcan vs UTMOST

TWAS for SLE using scPrediXcan

TWAS for SLE using scPrediXcan

1. TWAS for SLE using scPrediXcan and benchmarked its performance against the two canonical TWAS methods (pseudobulk and bulk; STAR Methods).
2. Compare the TWAS p-values

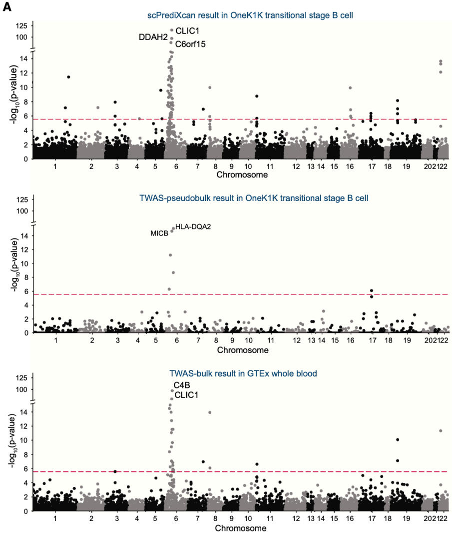


Figure: scPrediXcan's Performance

- scPrediXcan identified a greater number of candidate causal genes and explained more GWAS loci for SLE than the TWAS-pseudobulk and TWAS-bulk frameworks.
- scPrediXcan identifies 129 candidate causal genes from 24 different LD blocks among 1,703 pre-defined LD blocks,23 whereas TWAS-pseudobulk only identifies 11 candidate causal genes from 8 different LD blocks, and TWAS-bulk identifies 54 candidate causal genes from 14 different LD blocks.

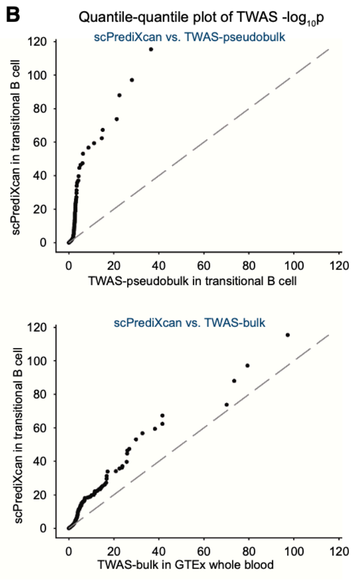


Figure: scPrediXcan's Performance

- scPrediXcan also pinpoints significant genes on other chromosomes that the other frameworks miss.

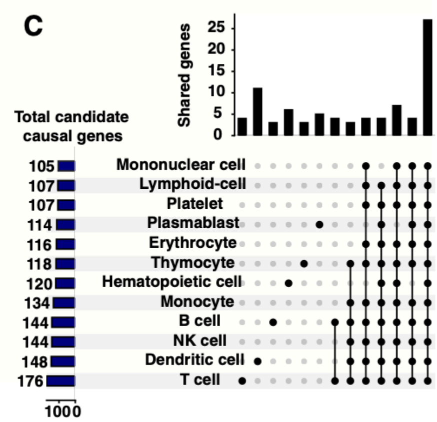


Figure: scPrediXcan's Performance

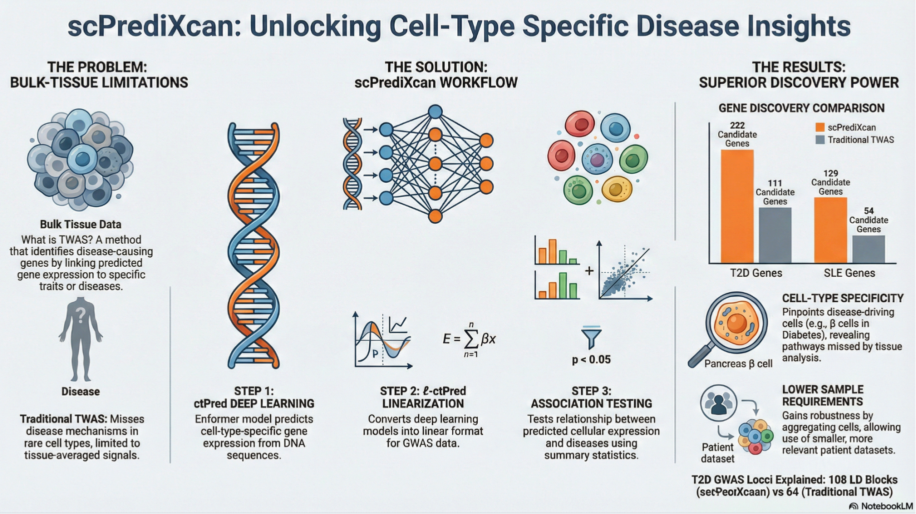
- While most TWAS hits are shared by different cell types, a few are cell type enriched.
- Among 243 TWAS hits from the 12 immune cell types, 27 (11.1%) genes are shared in all cell types, 205 (84.3%) genes are shared in at least two cell types, and 38 (15.6%) genes yield significance in a single cell type.

Conclusion

- The authors identified several potential driver genes for SLE among the cell-type-specific and cell-type-enriched associations that were overlooked in the bulk TWAS analysis.
- scPrediXcan can connect GWAS loci with genes known to have roles in SLE and other diseases but that had been missed by canonical TWAS and other expression-based integrative approaches.

Discussion

Discussion



scPrediXcan: Unlocking Cell-Type Specific Disease Insights
THE PROBLEM: BULK-TISSUE LIMITATIONS
Bulk Tissue Data
What is TWAS? A method that identifies disease-causing genes by linking predicted gene expression to specific traits or diseases.
Disease
Traditional TWAS: Misses disease mechanisms in rare cell types, limited to tissue-averaged signals.
THE SOLUTION: scPrediXcan WORKFLOW
STEP 1: ctPred DEEP LEARNING
Enformer model predicts cell-type-specific gene expression from DNA sequences.
$E = \sum_{n=1}^{n} \beta x$
STEP 2: ℓ-ctPred LINEARIZATION
Converts deep learning models into linear format for GWAS data.
p < 0.05
STEP 3: ASSOCIATION TESTING
Tests relationship between predicted cellular expression and diseases using summary statistics.
THE RESULTS: SUPERIOR DISCOVERY POWER
GENE DISCOVERY COMPARISON
scPrediXcan
Traditional TWAS
222 Candidate Genes
111 Candidate Genes
129 Candidate Genes
54 Candidate Genes
T2D Genes
SLE Genes
CELL-TYPE SPECIFICITY
Pinpoints disease-driving cells (e.g., β cells in Diabetes), revealing pathways missed by tissue analysis.
Pancreas β cell
LOWER SAMPLE REQUIREMENTS
Gains robustness by aggregating cells, allowing use of smaller, more relevant patient datasets.
Patient dataset
T2D GWAS Locci Explained: 108 LD Blocks (secPeoıXcaan) vs 64 (Traditional TWAS)
NotebookLM

References I

[1] Zhou, Y. *et al.* scPrediXcan integrates deep learning methods and single-cell data into a cell-type-specific transcriptome-wide association study framework. *Cell Genomics* **5**(5), 100875 (2025). doi:10.1016/j.xgen.2025.100875.

[2] Avsec, Ž., Agarwal, V., Visentin, D. *et al.* Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods* **18**, 1196–1203 (2021). doi:10.1038/s41592-021-01252-x.

The pictures other than the ones in the paper, with resource from the Internet, LaTeX diagrams, are generated by NotebookLM.

Thank you!

Appendix

Preprocess the data

- The dataset are randomly divided by chromosomes into training, validation, and test sets
- We used Enformer to predict the epigenomic features surrounding the transcription start site (TSS) of each gene. i.e., one gene one 896×5313 matrix
- We averaged the central four bins (447-450th) as the local regions of gene TSS into a linear 1 by 5,315 vector as the final epigenomic representation of the gene. The central four bins were determined empirically. With a total of 896 bins, bins 448 and 449 are central.
- For each cell-type—specific gene-by-individual count matrix at pseudobulk level, we averaged the read counts of each gene across individuals, ranked the averaged counts for each gene, and converted the ranks into percentiles ranging from 0 to 1. These percentile values were used as the target outputs for the ctPred model
- In our anaysis, we used the genotype data of Geuvadis individuals from 1000G Genome project.

Train the ctPred

	gene_name	chromo	0	1	2	3	4	5	6	7
0	ATAD3B	chr1	4.067672	3.647823	3.452344	1.299412	3.855479	5.222215	2.045481	3.731929
1	PRDM16	chr1	0.498504	1.675415	2.539938	0.854715	1.221890	1.847521	0.682262	1.584615
2	PEX10	chr1	10.101708	9.151280	8.493706	2.824118	9.172377	12.190944	4.258388	8.590059
3	PEX14	chr1	3.370678	2.529378	2.635246	1.991302	3.375057	4.478309	1.939699	2.746556
4	PLCH2	chr1	0.605982	0.576329	1.703161	0.194556	0.332990	0.525577	0.195734	0.522762
...	...	...	...	...	...	...	...	...	...	...
19015	CDY1	chrY	0.061454	0.053432	0.042596	0.143704	0.066984	0.065173	0.089574	0.071284
19016	TSPY4	chrY	0.006602	0.008855	0.014377	0.004839	0.007068	0.006743	0.003853	0.004656
19017	RBMY1J	chrY	4.402741	2.632909	1.950478	2.554271	4.028593	5.287105	2.040063	2.882508
19018	RBMY1B	chrY	3.020573	1.685974	1.149323	1.760193	3.046081	4.016865	1.424957	2.056236
19019	RBMY1A1	chrY	4.257495	2.799338	2.017697	2.541247	4.215118	5.173813	2.341953	3.170462

19020 rows × 5316 columns

Figure: Training data, with the last column the population-average expression level in certain cell types, distinct models are trained for each cell type, they did not say it specifically but it seems that the same 5313 features are used for all cell type

Train the ctPred

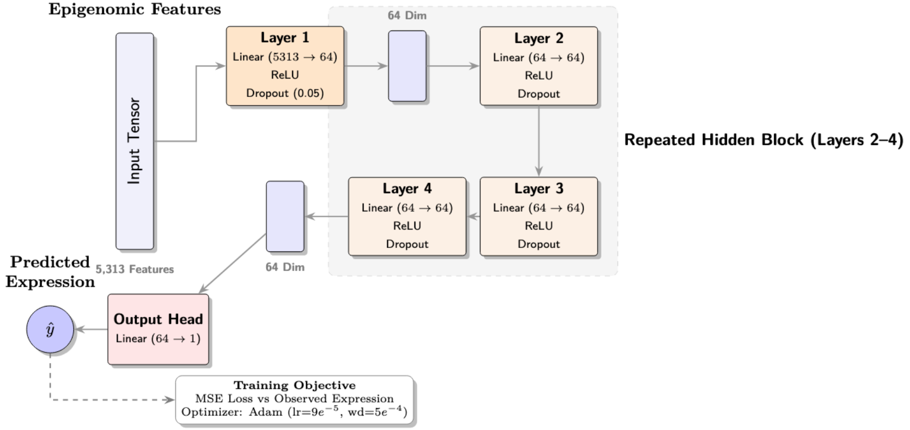


Figure: MLP architecture

Train the ℓ-ctPred

- Training data: 462 European individuals from 1000G project
- The stark difference in the number of converging genes arises because the canonical TWAS prediction model is based on observed data, whereas l-ctPred relies on in silico ctPred-predicted gene expression data. The in silico data contain only the genetic component of expression, while the observed gene expression includes both genetic and environmental components, making l-ctPred less vulnerable to the noise and sparsity present in observed scRNA-seq data.
- Analysis: 40 cell types from T2D and OneK1K datasets

Train the PEN

- PEN: SNP $\rightarrow$ RNA(expression)
- Trained on same observed single-cell data as ctPred
- Failed to converge for the majority of genes, predict for only 3.5% of expressed genes
- The metric $m1 = \pi_1 \times \#$ genes stands for "true positive genes", where π_1 is estimated from q-value method.

Benchmark the scPrediXcan

- **TWAS-pseudobulk**: train cis-SNP $\rightarrow$ expression predictors using cell-type pseudobulk scRNA-seq (gene$\times$individual); test gene–trait association at cell-type level.
- **TWAS-bulk**: train cis-SNP $\rightarrow$ expression predictors using bulk tissue RNA-seq (e.g., GTEx); test gene–trait association at tissue level.
- **UTMOST**: cross-tissue TWAS that jointly learns SNP $\rightarrow$ expression effects across multiple tissues to borrow strength and improve power.

Enformer's Architecture

- **RConv (residual conv block)**: a skip connection adds the input back to the conv output, helping gradients flow and preventing information loss as depth increases.
- **BatchNorm**: normalizes activations using batch statistics to stabilize training and speed up convergence (commonly used in convolutional stacks).
- **GELU**: smooth nonlinearity (used in Transformers) that improves expressiveness compared with ReLU for many NLP/genomics models.
- **LayerNorm**: normalizes per token/position across channels, making Transformer attention blocks train stably and less sensitive to batch size.
- **Dropout**: randomly zeros activations during training to reduce overfitting and improve generalization.
- **Softplus**: smooth, always-positive output activation (useful when predicting non-negative quantities such as coverage/count-like tracks).

Relative Positional Encoding

- **Idea**: instead of giving each position an absolute index, encode *relative distance* $(i - j)$ inside attention so the model learns "how far" two bases are and whether one is upstream/downstream.
- **How it enters attention (Transformer-XL style)**: add a relative-position term R_{ij} to the query–key interaction, so attention scores depend on both content similarity and distance.
- **Why for genomics**: regulatory effects are distance-dependent (e.g., promoter–enhancer), and relative encoding generalizes to shifts and long-range interactions better than absolute positions.
- **Enformer detail**: represent distance using a bank of basis functions (exponential decay, local "central mask", and gamma-shaped kernels) and include both **symmetric** $f(|i - j|)$ and **directional** $\mathrm{sign}(i - j)\, f(|i - j|)$ features.